



CLASSIFYING IRISH TUNE TYPES FROM ABC NOTATION WITH A CHARACTER-LEVEL LSTM

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ABSTRACT

Irish traditional tune types (reels, jigs, hornpipes, etc.) have been classified using hand-crafted symbolic features or audio embeddings of synthesized scores; neural sequence models trained on the same ABC-notation corpora have so far been used only for tune generation, not classification. This work trains a character-level LSTM directly on raw ABC-notation transcriptions from The Session to classify tune type across twelve categories, comparing it against a support-vector-machine baseline built on music21-derived pitch, interval, and rhythm histograms, both evaluated on a piece-level split to avoid leakage between settings of the same tune.

INTRODUCTION

Irish tune types are set apart mainly by meter and rhythmic feel rather than pitch content. Prior approaches to automatic classification fall into distinct groups, and a real gap remains for a neural sequence model trained end-to-end and directly on raw ABC text.

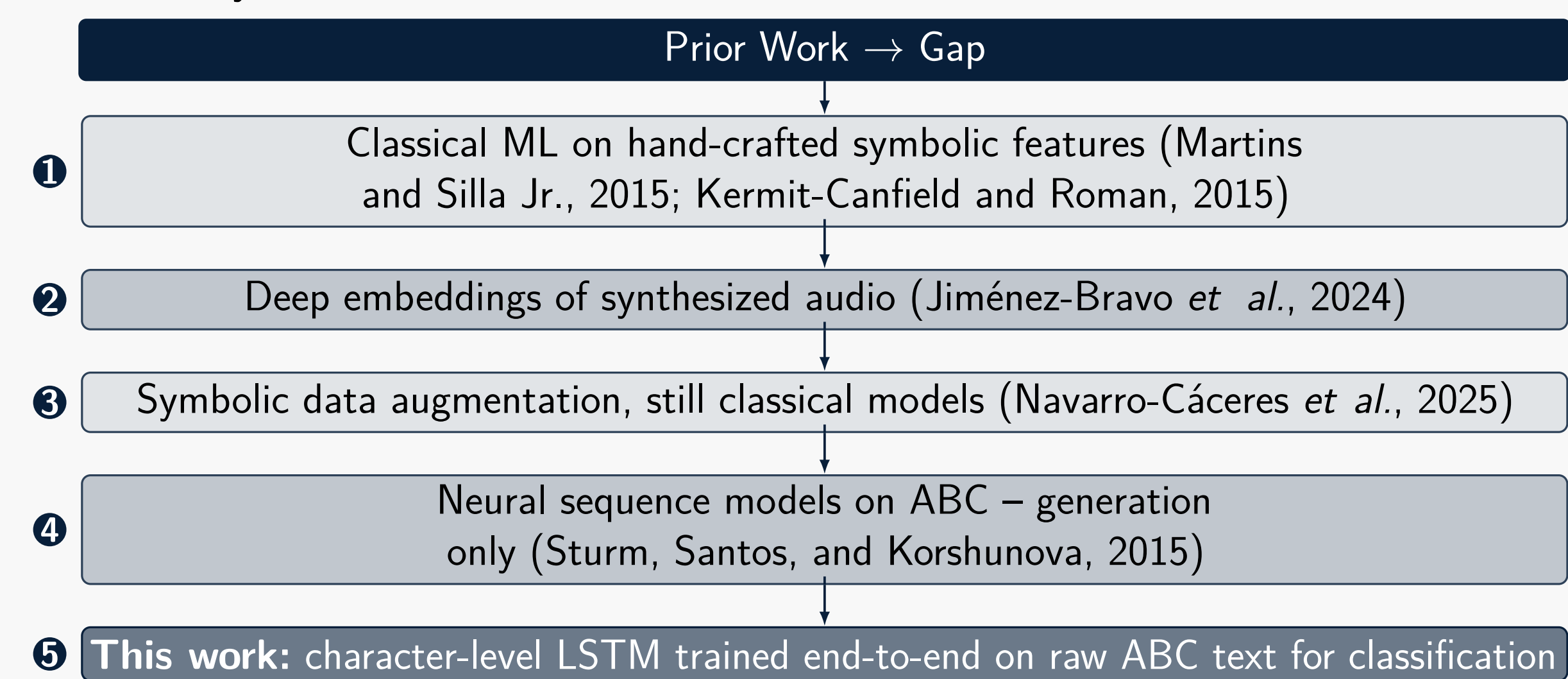


Figure 1 – Prior approaches to Irish tune-type/genre classification and the remaining gap this work addresses.

DATA EXAMPLE

Raw ABC text, read directly by the LSTM character by character:

Reel (4/4): |:eAAB eABB|eAAB gedB|eAAB eABB|G2AB gedB:|

Jig (6/8): |:A2D DEG|A2E G2E|A2D DEG|AGE B2G|

The reel's straight 4/4 pulse and the jig's rolling 6/8 grouping are visible directly in the text the model sees.

ACKNOWLEDGEMENTS

Tunes and type labels are drawn from The Session, a community archive of Irish traditional music.

METHOD

Data. 55,352 settings / 23,300 unique tunes / 12 types from The Session, split 70/15/15 at the tune level (not the setting level) to avoid leakage between settings of the same tune.

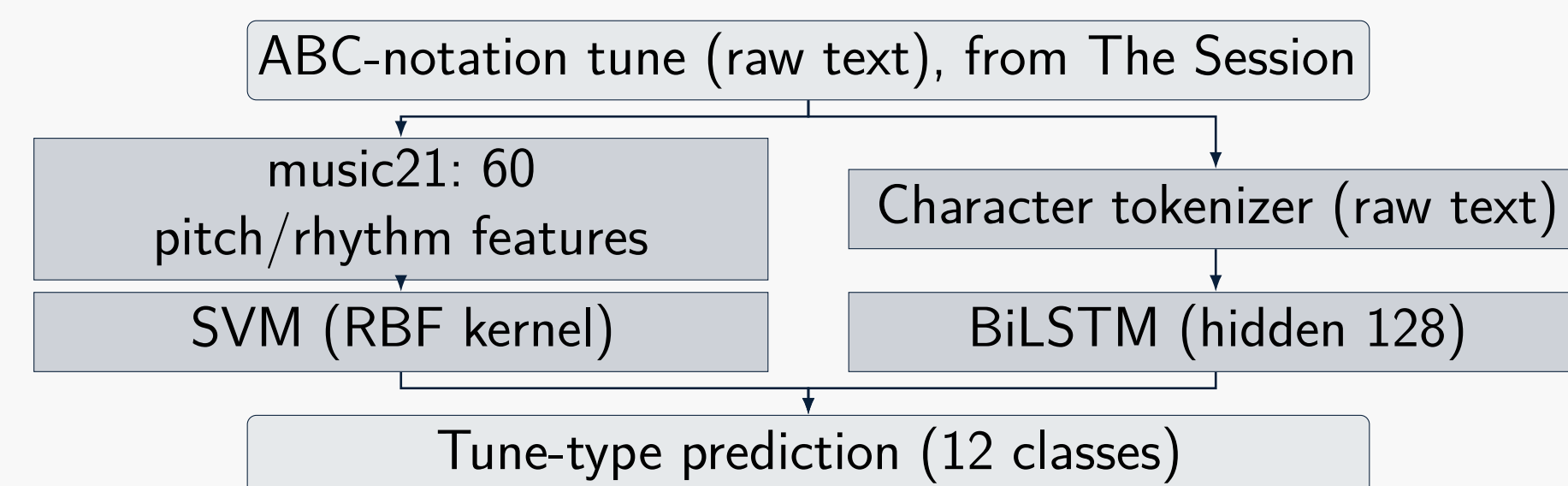


Figure 2 – The two classification pipelines compared: a feature-engineered SVM baseline and an end-to-end LSTM trained directly on raw ABC text.

Training. AdamW (Loshchilov and Hutter, 2019) with warmup and cosine decay, gradient clipping, dropout 0.3, weight decay 10^{-4} , early stopping on validation loss.

ABLATION STUDY

Class weighting, regularization, and transposition augmentation were evaluated one at a time on the LSTM.

Class w.	Reg.	Aug.	Acc.	Macro-F1
✓	✗	✗	77.5%	0.650
✓	✗	✓	76.5%	0.628
✓	✓	✗	78.4%	0.653
✗	✓	✗	83.9%	0.692

Table 1 – Ablation over the LSTM training recipe: class-weighted loss, regularization (dropout + weight decay), and transposition augmentation.

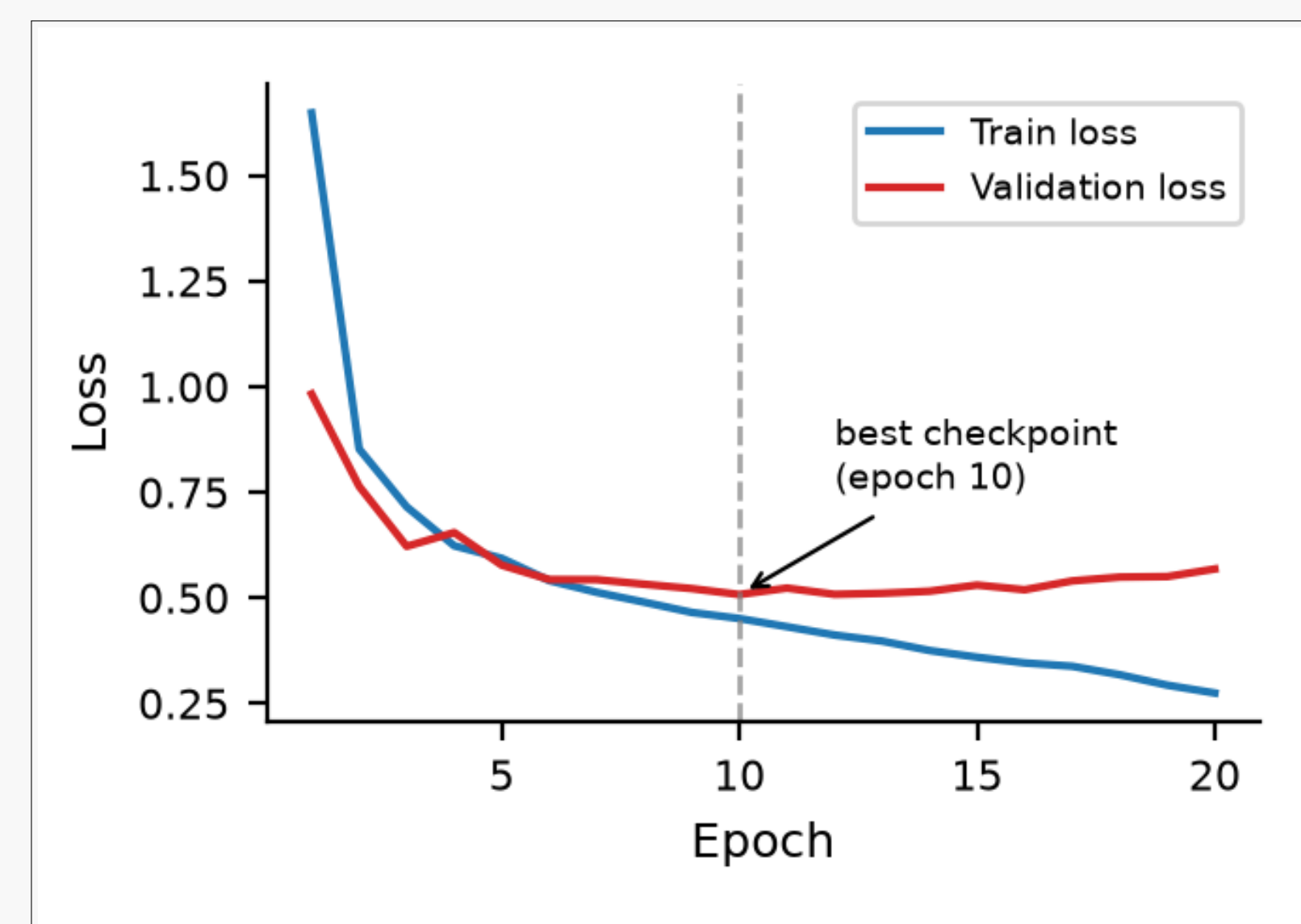


Figure 3 – LSTM train/validation loss by epoch. Early stopping restores the checkpoint at epoch 10.

RESULTS

Without class-weighted training, the LSTM reaches 83.9% test accuracy, exceeding the SVM baseline (82.7%), but the SVM retains a higher macro-F1 (0.739 vs. 0.692) – a trade-off between accuracy and balanced performance on metrically similar minority classes.

Model	Acc.	Macro-F1
SVM (RBF)	82.7%	0.739
LSTM (class-weighted)	78.4%	0.653
LSTM (unweighted)	83.9%	0.692

Table 2 – Test-set results, SVM baseline vs. LSTM under class-weighted/unweighted training.

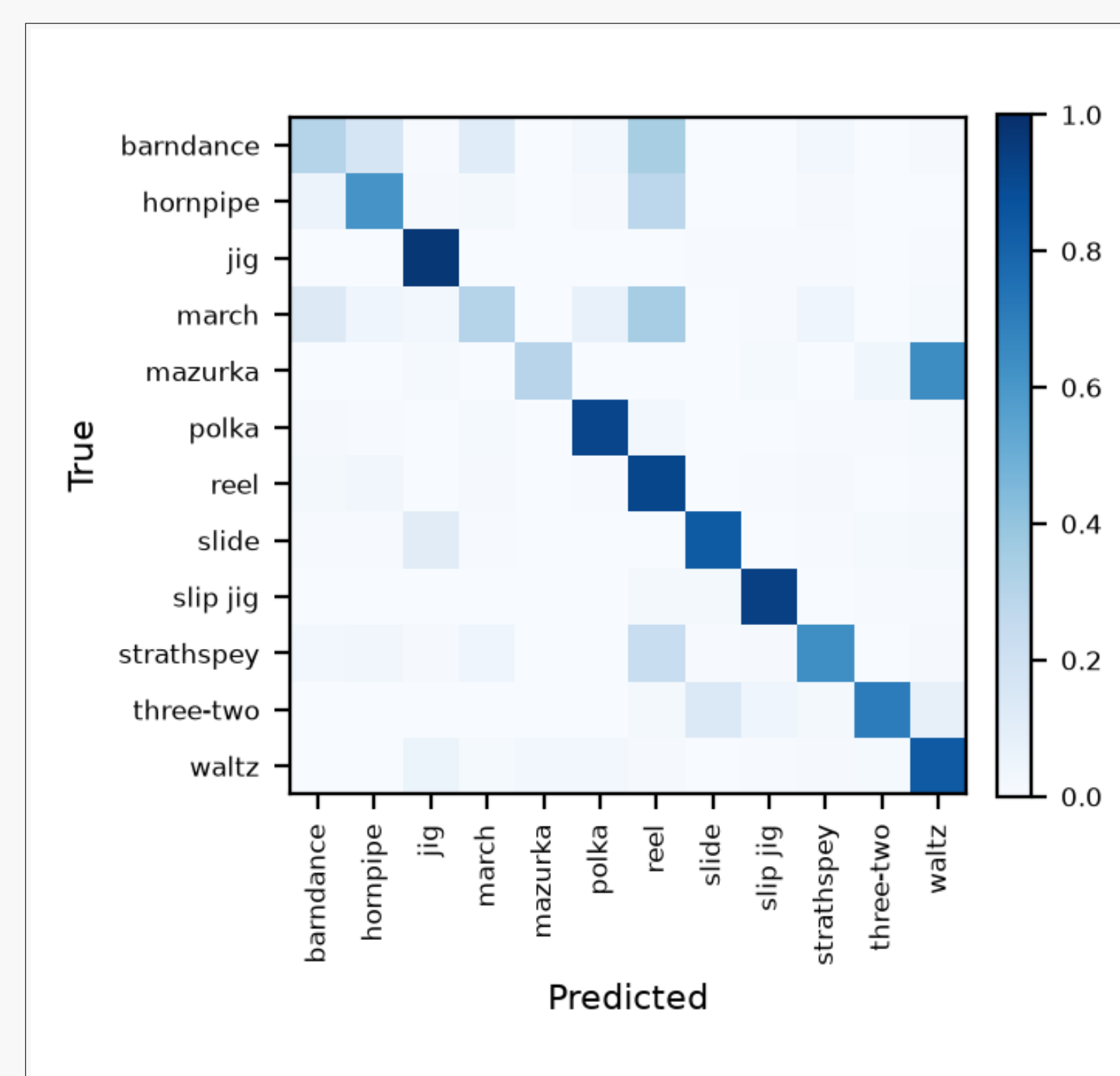
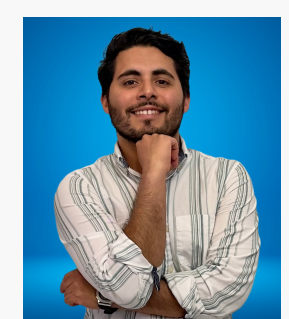


Figure 4 – Confusion matrix of the unweighted LSTM (row-normalized). Reel, hornpipe, and barndance (all 4/4) are confused with each other; jig's 6/8 feel is unambiguous.

CONCLUSION

A character-level LSTM trained directly on raw ABC-notation text can match or exceed a feature-engineered SVM baseline in accuracy, without audio synthesis or hand-crafted features, though it does not resolve the class-imbalance problem the same baseline family already exhibited. Future work: cross-validation, comparison against recent symbolic-augmentation approaches (Navarro-Cáceres *et al.*, 2025), and dedicated class-imbalance techniques for sequence models.

PRESENTER



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CODE & RESULTS

